Statistical Methods For Market Research

Introduction and Chapter 1

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Course Objectives

- Designing a market research project
- Gaining experience in using statistical software packages
- Knowing how to interpret output from statistical software and drawing appropriate conclusions.

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University of London Examination

- This course will make every effort to prepare students to do well in the UoL exam
- Therefore, the course will follow the course material on the UoL website

- If you are not taking the UoL exam, the material in this course is still valuable

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Use of English

- The class will be taught in English, and the homework and exams will also be in English

- I will try to talk clearly and slowly so all can follow. **PLEASE** speak up if you have trouble understanding

- I have no problem repeating material in Spanish if the language becomes a barrier

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Grading

- Homeworks- 25 points. 5 short homeworks to keep you on track and practicing the course material. If you use AI, you won't be prepared for the Mock. In groups of up to three people.

- Project- 20 points. "Individual" Case Study of UoL. Students taking Exam should do it on their own, while other students can work in groups of up to three. Grades will be adjusted to reflect the effort of working alone.

- Exam(Parcial)- 45 points. Will be a "Mock" of the UoL Exam

- Class Participation- 10 points. Speak, Pay Attention

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Artificial Intelligence Policy

- AI should be used as a tool, not your replacement
- AI is allowed to help with writing, and coding when necessary
- If it becomes clear to me that AI use is rampant, I will ask for the version of your project before using AI.
- It will become clear to me who uses AI when I grade the exam which will be with paper and pencil :).

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Classroom Policy

- 75% required like in all UM courses
- Please stay focused during the class. If the material is heavy, I will give a 5 min break to regain focus
- If electronic devices become a distraction I will ask you to go back to paper and pencil.

PLEASE ASK QUESTIONS!!!!!!!!!!!!!!!

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Learning Outcomes

- define a market research problem and create an appropriate research design
- perform independent data analysis in a market research setting
- determine which statistical method is appropriate in a given situation and be able to discuss the merits and limitations of a particular method
- use statistical software to analyze datasets and be able to interpret output
- draw appropriate conclusions following empirical analysis and use to form the basis of managerial decision-making
- demonstrate greater commercial awareness.

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WHAT IS MARKETING???

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What is Marketing?

- Marketing is the activity, set of institutions, and processes for creating, communicating, delivering, and exchanging offerings that have value for customers, clients, partners, and society at large **(American Marketing Association)**

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Distinct Roles In Marketing

Market Researcher

Responsible for designing and crafting high-quality research and providing relevant information, helping to:

- Describe the nature and scope of customer groups
- Understand the nature of forces that shape customer groups
- Understand the nature of forces that shape the marketer's ability to satisfy target customer group
- Test individual and interactive variables that shape customer experiences
- Monitor and reflect upon past success and failures in marketing decisions

Decision Maker

Makes the final decision



Uneasy lies the head that wears a crown.

Henry IV, Part II

Shakespeare

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What is the Goal?

A decision-maker must optimize the

Strategic Market Position

taking into account:

- Customer Targeting
- Competitors Activities
- The Company
- Customer Research

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What Decisions Can Be Made?



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The Marketing Mix

Product – a tangible good or an intangible service for which decisions about the design, development, testing of a new product, improvement of existing ones, packaging etc. are taken.

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The Marketing Mix

Price – an indicator of quality, where pricing decisions will impact profits and the marketing strategy. Price discrimination (charging different prices to different consumers) could be applied, which would require information on different consumers' price elasticities of demand which quantifies price sensitivities.

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The Marketing Mix

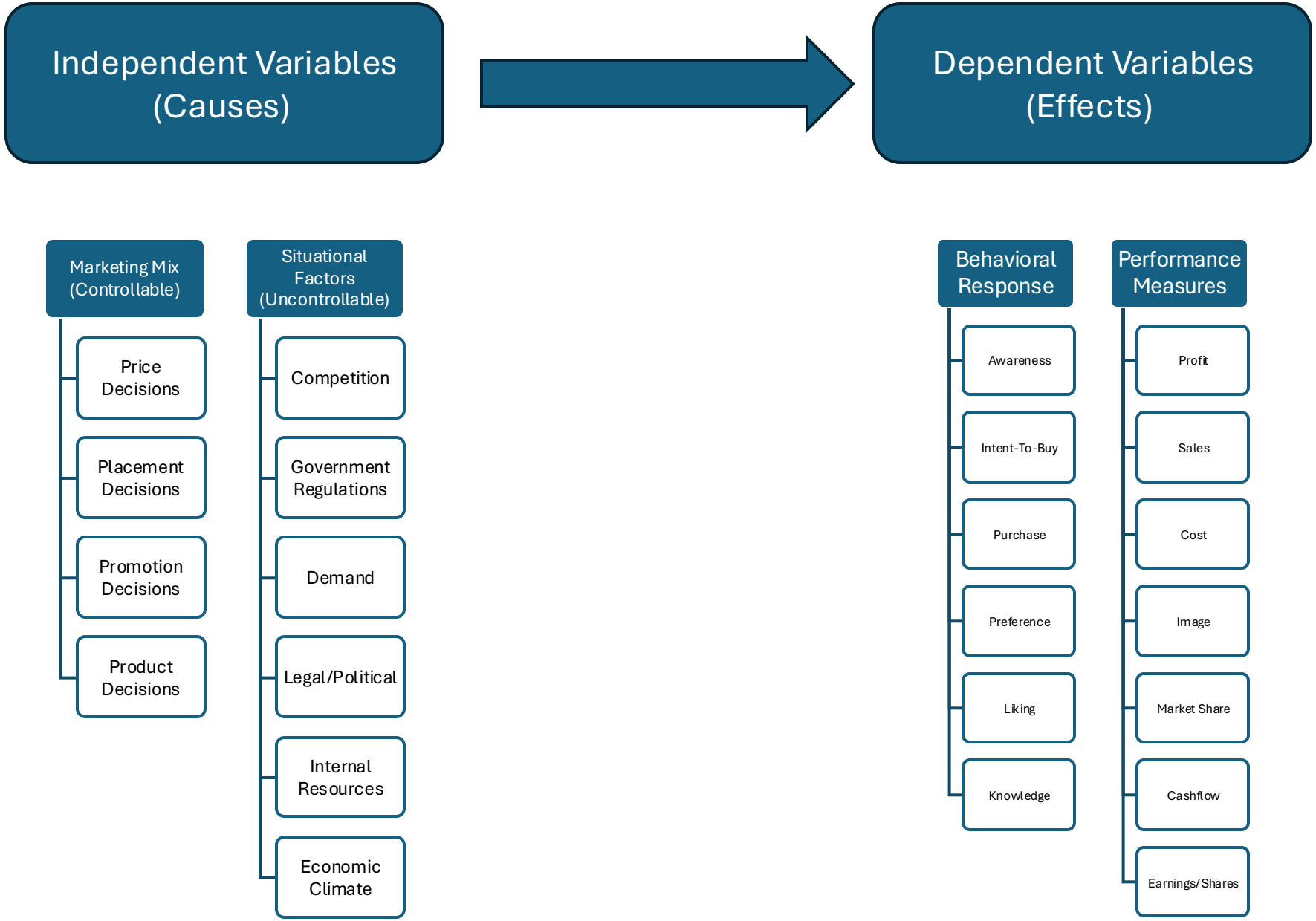
Placement – ensuring a firm positions its product in the right place, at the right time, at the right price. Even e-commerce firms need to engage potential customers with appropriate web placement.

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The Marketing Mix

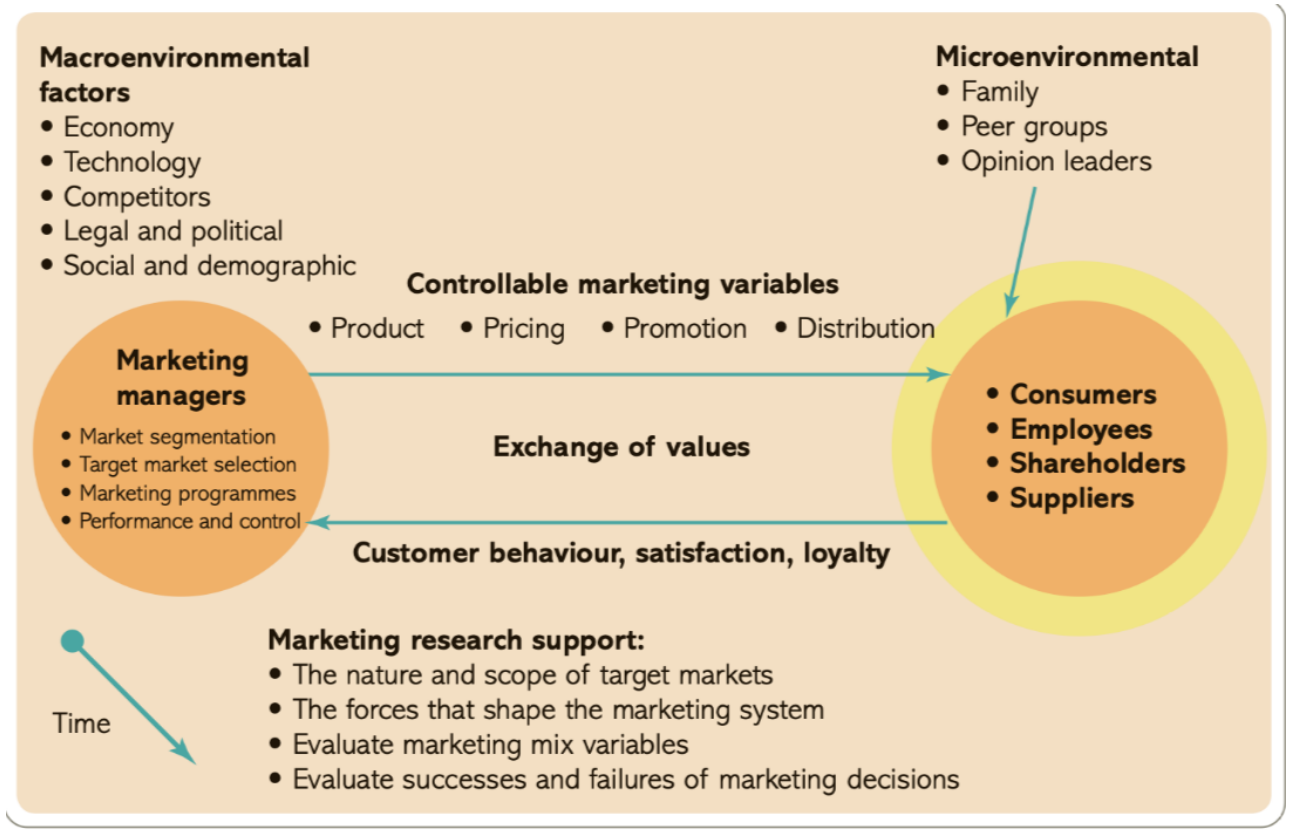
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The Marketing System



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The Role of The Researcher

- Describe the nature and scope of customer groups
- Understand the nature of forces that shape customer groups
- Understand the nature of forces that shape the marketer's ability to satisfy targeted customer groups
- Test individual and interactive variables that shape customer experiences
- Monitor and reflect upon past success and failures in marketing decisions

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Evolution of Market Research

- The concepts have been applied since ancient times

1820s- First opinion polls in the US

1890s- Wide use of questionnaires to gauge consumer's opinions of advertising

1910s to 1930s – Evolution of market research into mainstream and commercial use

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Definitions of Market Research

Market research, which includes social and opinion research, is the systematic gathering and interpretation of information about individuals or organisations using the statistical and analytical methods and techniques of the applied sciences to gain insight or support decision making. The identity of respondents will not be revealed to the user of the information without explicit consent and no sales approach will be made to them as a direct result of their having provided information

(ESOMAR)

Marketing research is the function that links the consumer, customer, and public to the marketer through information – information used to identify and define marketing opportunities and problems; generate, refine, and evaluate marketing actions; monitor marketing performance; and improve understanding of marketing as a process. Marketing research specifies the information required to address these issues, designs the method for collecting information, manages and implements the data collection process, analyzes the results, and communicates the findings and their implications.

(American Marketing Association)

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Classification of Market Research

Problem Identification Research

- Market research undertaken to help identify problems which are not necessarily obvious in the present, yet exist or are likely to occur in the future

Problem-Solving Research

- Market research undertaken to help solve specific marketing problems.

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The Marketing Research Process (Introduction)

1. Problem Definition
2. Developing a Research Approach
3. Research Design
4. Fieldwork or Data Collection
5. Data Analysis
6. Communicating Findings

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1. Problem Definition

- Can be the most challenging stage in a research project due to the ambiguous nature

Might Include:

- Discussions with Decision Makers
- In Depth Interviews
- Collection and Analysis of readily available information

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2. Developing A Research Approach

- Need to identify factors that influence a research design

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3. Research Design

- Framework for conducting a Market Research Project
- Details the necessary procedures required and how the information will be obtained
- Defining the variables to be measured and the designing appropriate scales can be a part of this(depending on the question or procedure)

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4.Fieldwork or Data Collection

1. Always start with a review of existing Secondary Data

Secondary Data are data collected for some other purpoe than the problem at hand.

Secondary Data might have enough information to complete the research process

If it is not enough to complete the research process, reviewing secondary data avoids replication of work and can help guide data collection(sampling, techniques, variables

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4.Fieldwork or Data Collection

2. Identify and Select Individuals for primary research

- "Representativeness" is more important in quantitative research and therefore a good sampling method would be required

- In qualitative research "Representativeness" is less important than the quality of the individuals and their answers.

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4.Fieldwork or Data Collection

- Research can include quantitative and qualitative data collection
- Both can be done at the same time or one before another.

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5. Data Analysis

- The most important step is to convert collected data and information into data that can be analyzed

- Data cleaning, editing and verification are necessary part of a good analysis.

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6. Communicating Findings

- Before: A written report that address the specific research questions identified, describes the approach, research design, data collection and data analysis procedures adopted and presents the results and major findings

- Now: Make it concise

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6. Communicating Findings

- Due to limited manager's time, reports need to be concise and are more visual than before

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Defining The Problem

Of all the tasks in a marketing research project, none is more vital to the ultimate fulfilment of a client's needs than an accurate and adequate definition of the research problem. All the effort, time, and money spent from this point on will be wasted if the problem is misunderstood and ill-defined.

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Defining The Problem

Avoid Being Dr.House

What Does He Do?
Prescribes medicine before identifying the correct problem



Defining The Problem

- The first is to state the general problem and identify the specific components of the Marketing Research Problem .

- The first step is the **Research Brief,** where the buyer of the research communicates the perceived requirement

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The Research Brief

- Created at the beginning to clarify between all parties and to avoid future conflict



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Marketing Research Proposal

1.*Executive summary* – a concise overview of the whole proposal.

2. *Background* – background information about the firm and its industry more widely.

3. *Problem definition* – clarifying the problem to be addressed.

4. *Research objectives* – possible research hypotheses to be tested.

5. *Research design* – the research design to be employed (exploratory, descriptive or causal - outlined in the next block).

6. *Fieldwork/data collection* – the logistics of collecting any primary data, including the sampling method(s) to be used.

7. *Data analysis* – specification of the type(s) of statistical analysis to be conducted to investigate the research objectives.

8. *Reporting* – whether interim reports are to be produced, and the format of the final report.

9.*Cost and timetable* – an indication of the budget required and the expected timescale for the research.

10. *Research organization and researchers* – details of the individuals and the market research organization behind the proposal.

11. *Appendices* – any non-essential tables etc. can be relegated to an appendix.

12. *Agreement* – a formal document to be signed by all parties.



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Decision vs Research Problems

Marketing Decision Problem	Market Research Problem
Should a firm bring a new product to market?	Research consumer preferences and purchase intentions for the proposed new product.
Should an advertising campaign be changed?	Research the effectiveness of the current advertising campaign.
Should the price of the product be increased?	Research the price elasticity of demand and the impact on sales and profits of various price level changes.

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Limitations of Marketing Research

- Marketing research is not very good for innovating

"Some people say, ‘Give the customers what they want.’ But that’s not my approach. Our job is to figure out what they’re going to want before they do. I think Henry Ford once said, ‘If I’d asked customers what they wanted, they would have told me, “A faster horse!” People don’t know what they want until you show it to them. That’s why I never rely on market research. Our task is to read things that are not yet on the page"
-Steve Jobs, Founder Apple



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What Marketing Research Can do

- By carrying out research on the effectiveness of search-engine advertising, it found that it could reduce expenditure on certain keywords, such as ‘easyJet’, without reducing the number of visitors to the website.

- Created savings of £8 million in savings that was used in TV ads



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